Renal Function Tests

Urea, Creatinine, & Uric acid

College of Medicine - HMU

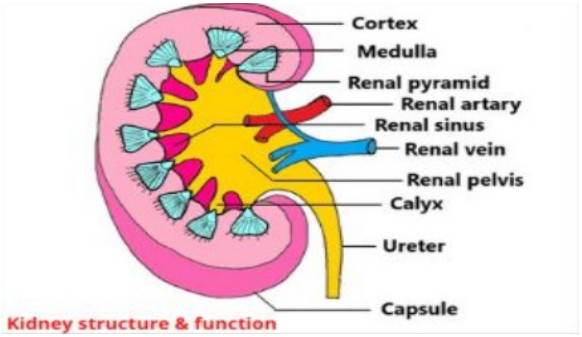
2nd year Medical Students

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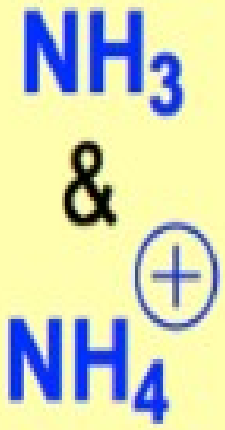


Learning Outcomes:

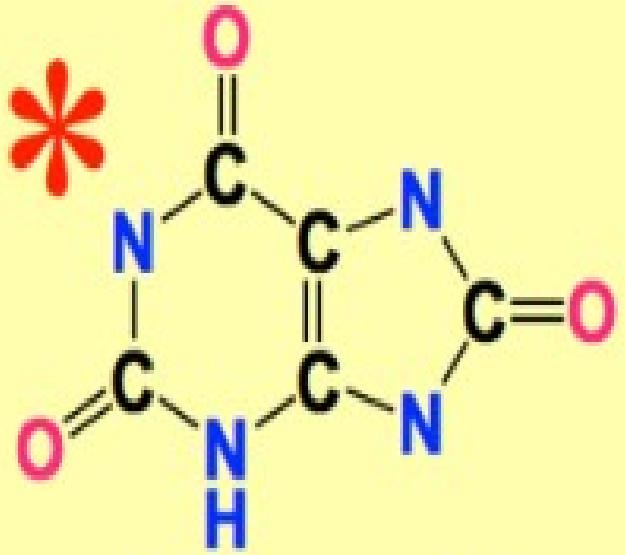
- Explain the biochemical origin of urea, creatinine, and uric acid.
- Describe the main renal handling and excretion of these nonprotein nitrogenous compounds.
- Interpret common causes of increased or decreased serum levels.
- State the basic laboratory principles used to estimate urea, creatinine, and uric acid.

Introduction:

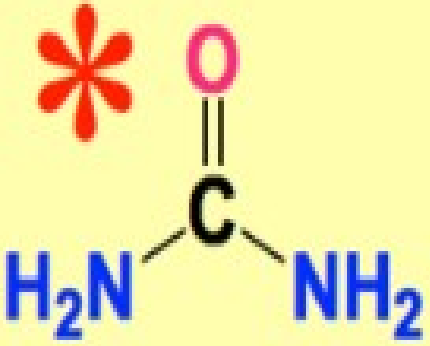
- Urea, creatinine and uric acid are three principle nonprotein nitrogenous substances (NPN).
- Urea and creatinine are derived mainly from protein metabolism, while uric acid is the end product of purine metabolism.



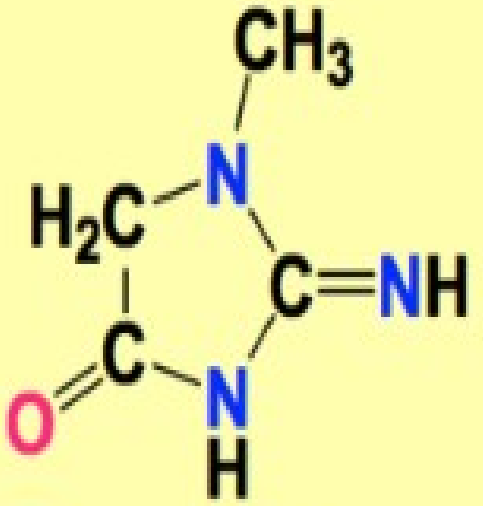
Ammonia



Uric acid



Urea



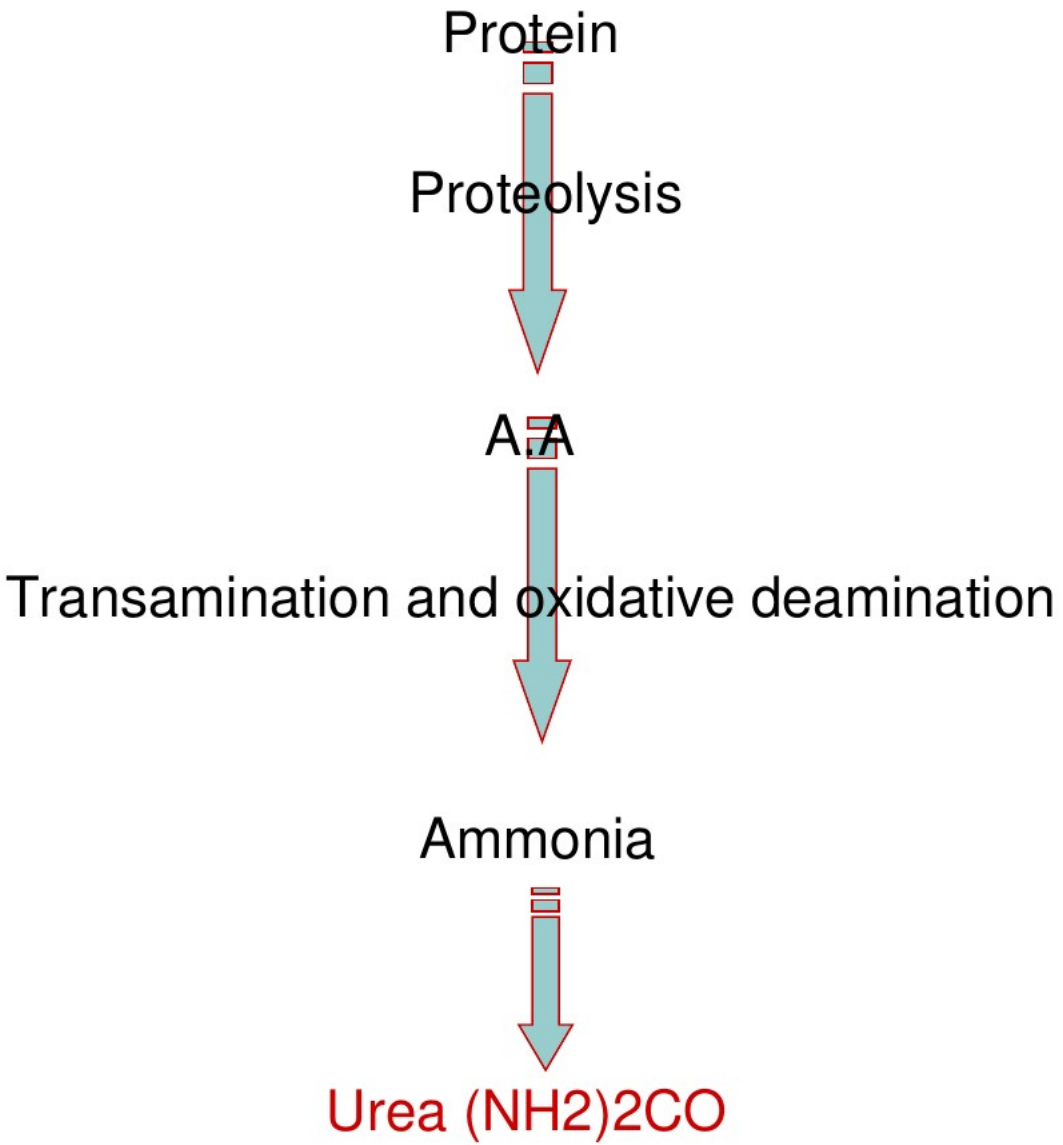
Creatinine

1- Urea:

- Urea is a relatively nontoxic substance made by the liver as a means of disposing of ammonia from protein metabolism.
- It accounts for about 90% of the nitrogen containing components of urine.
- Less amount of urea excreted through gut and sweat. While about 90% is excreted by kidney and renal excretion depend on renal blood flow, glomerular filtration rate, and rate of urine flow.

Urea, cont.,

- About 40-60% of urea filtrate by the glomeruli is reabsorbed in the collecting ducts,
- the amount of reabsorption varies inversely with rate of urine flow.


Protein
Proteolysis
A.A
Transamination and oxidative deamination
Ammonia
Urea (NH2)2CO

2- Creatinine:

- Creatinine is derived from creatine, which synthesized endogenously in liver and pancreas from 3 a. acids glycine, methionine and arginine.
- Creatine is reversibly phosphorylated in muscle to creatine phosphate, which is a high energy compound that can be used to maintain the intracellular level of ATP during muscular contraction.
- Creatine and creatine phosphate spontaneously cyclize to form creatinine, which is excreted in the urine.

3- Uric Acid:

- Uric acid is the end product of purine metabolism (adenine & guanine),
- which are present in nucleic acids (DNA & RNA).

Mechanism of Uric acid Metabolism

DNA & RNA

↓

Purine

↓

Hypoxanthine

↓xanthine oxidase

Xanthine

↓xanthine oxidase

Uric acid

25-30%
GIT

70-75%
Kidney

Normal Range:

BU (blood urea) (8-40) mg/dl

s. creatinine (0.6-1.2)mg/dl

s. uric acid (3-7)mg/dl



Clinical Significance:

A- Increased blood Urea and Creatinine:

1. Increased production: due to increased meat intake, and a very large muscle mass increase level of creatinine.

2. Impaired kidney function may be prerenal, renal, or postrenal.

i- Prerenal: dehydration, hemorrhage, shock, and congestive heart failure.

ii- Renal: chronic nephritis, and acute tubular necrosis.

iii- Postrenal: any obstruction to urine flow (stone, tumour, enlarged prostate).

B- Decreased blood Urea and Creatinine:

Decreased blood urea caused by

1. Decreased synthesis like in liver disease and decreased protein intake in diet.

2. Increased excretion like in nephrotic syndrome and pregnancy.

Decreased creatinine found in advanced liver disease, paralysis or muscular dystrophy.

C- Increased level of serum uric acid:

1- Increased overproduction: due to increased purine synthesis (ribose-5-p overproduction), and increased breakdown of nucleoprotein (burns, psoriasis).

2- Reduced excretion:

(renal failure, thiazide diuretics).

3- Diet rich within purine like liver, meat, and fish.

Psoriasis





Clinical Effects of Prolonged Hyperuricemia:

1- Renal stones.

2- Renal and vascular diseases.

3- Gouty arthritis.

Gouty Patients





D- Decreased level of serum uric acid:

Decreased level of uric acid caused by xanthine oxidase deficiency, which results from a genetic defect or to severe liver damage.

Principle:

Urea estimated by Nessler's method:-

The urease enzyme hydrolyses urea to ammonium carbonate which on treatment with Nessler's reagent produces yellowish orange color.

$CO(NH_2)_2 + H_2O \xrightarrow{\text{urease } 37C} (NH_4)_2CO_3$

$(NH_4)_2CO_3 + 2NaOH \longrightarrow Na_2CO_3 + 2NH_3 + 2H_2O$

$HgI_2 + 2KI + 2NH_3 \longrightarrow NH_2Hg_2I_3 + 2KI + 2NH_4I$

Yellow-orange color

Creatinine estimation is based on the jaffe's method. Creatinine in alkaline condition reacts with picric acid to form creatinine picrate, which has a reddish yellow color.

Uric acid is enzymatically determined using uricase enzyme, uric acid oxidized by oxygen to produce hydrogen peroxide, allantoin, and carbon dioxide.
Uric acid + phosphotungstic → Allantoin+ CO_2+ tungsten blue

Lab Procedure

- Incubation: **Mix 100 µL serum + 0.5 mL reagent** and **leave for 2 mins at room temperature** until it turns reddish yellow.
- Measurement: Zero the spectrophotometer with a reagent blank, then read the standard and patient sample at 578 nm.

$$\text{Concentration of Sample} = \frac{\text{Absorbance of Sample}}{\text{Absorbance of Standard}} \times \text{Concentration of st.}$$

References:

- Tietz Fundamentals of Clinical Chemistry and Molecular Diagnostics. Elsevier.
- Henry's Clinical Diagnosis and Management by Laboratory Methods. Elsevier.
- Tietz Textbook of Laboratory Medicine. Elsevier.
- KDIGO resources for evaluation of kidney function and use of estimated GFR.

Thank You

Questions?